

Josiah deGrasse

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Human Factors engineer and AI product builder at Tufts. I turn messy real-world workflows into tools people can actually use - especially when AI has to be explainable, reviewable, and easy to hand off.

EDUCATION

Tufts University

Medford, MA

Bachelor of Science in Human Factors Engineering

May 2026

RELEVANT EXPERIENCE

New Food Innovation

Sheffield, United Kingdom

AI Product Engineer

Jan 2026 - Present

- Hired by the project sponsor to turn a Tufts capstone into a working sensory-intelligence platform for food-development teams, owning design, UX, full-stack build, testing, and deployment.
- Built the core workflow from scratch: instrument-data import, AI-assisted CSV mapping, panelist assignments, configurable surveys, sensory analysis, concept testing, and branded PDF/Excel reports.
- Created the **ISSF decision framework**, translating e-tongue, GC-MS/GC-O, chemical, and human-panel evidence into explainable GO/TWEAK/STOP recommendations with confidence scores, gates, and human approval.
- Added AI where it made the product faster but still reviewable: packaging concepts, survey-question drafts, report narratives, food-type detection, and commercialization summaries.
- Shipped a React, TypeScript, and Supabase product with role-based admin/panelist flows, provenance tracking, PDF/XLSX exports, and a 79-test automated suite.
- Made the system usable for food scientists and commercial teams by turning complex sensory methodology into guided workflows instead of spreadsheets and hand-written reports.

Red Hat OpenShift AI - Senior Capstone

Somerville, MA

Product Designer

Jan 2026 - May 2026

- Designed and tested two versions of an enterprise AI troubleshooting assistant - one RAG-based and one rule-based - to learn when users trusted AI help and when they wanted more control.
- Ran user interviews and stakeholder sessions, then built Figma prototypes around real OpenShift workflows including onboarding, hardware setup, YAML diffs, and deployment handoffs.
- Found where users lost trust in the assistant and redesigned the experience around visibility, source grounding, rollback, and clearer human control.

Helfrich Brothers Boiler Works

Lawrence, MA

Mechanical Engineer

Jun 2024 - Aug 2024

- Designed production fixtures for robotic welders and CNC machinery that increased boiler-head throughput from 3 heads per 3-hour run to 8 heads per 45 minutes.
- Modeled 1,000+ SolidWorks components and digital twins to support shop-floor decisions, documentation, BOM accuracy, and rework reduction.

LEADERSHIP, TEACHING & COMMUNITY

Tufts University Men's Lacrosse Team

Medford, MA

NCAA Student-Athlete

Sept 2022 - May 2026

- Contributed to three consecutive NCAA Division III National Championships while balancing 30+ hours/week of training, games, travel, and academics.
- Founded a weekly team newspaper and helped create peer-support pathways that made it easier for teammates to communicate, ask for help, and stay connected.

Youth & Community Lacrosse Coaching

Ridgefield, CT & Bronx, NY

Volunteer Coach

2019 - Present

- Coach youth athletes in Ridgefield and with Bronx Lacrosse, expanding access to the sport while mentoring younger players across skill levels and backgrounds.
- Fundraise annually for HEADstrong, supporting families affected by pediatric cancer through a lacrosse-based giving network.

SKILLS

AI/Product: Claude/LLM workflows, prompt design, AI-assisted reporting, human-in-the-loop review, eval thinking, workflow automation

Technical: React, TypeScript, Python, FastAPI, Node.js, Supabase/PostgreSQL, SQL, REST APIs, Git, Figma, R

Design/Research: Human factors research, usability testing, ACTA, prototyping, stakeholder interviews, SolidWorks, CAD, 3D printing

Interests: Sewing, sourdough baking, snowboarding, surfing, backgammon, music technology